

EXPERIMENT 12

AIM: Write a Program to implement Broadcast tree by taking subnet of hosts.

SOLUTION

Algorithm: Broadcast Tree Using Cost Matrix

1. Start the program.
2. Read the number of hosts in the subnet.
3. Read the cost matrix, where 0 indicates no direct connection and positive values indicate link costs.
4. Select Host 1 as the root and mark it as visited.
5. Find the minimum-cost connection between a visited host and an unvisited host.
6. Add the selected connection to the broadcast tree and mark the new host as visited.
7. Repeat Steps 5 and 6 until all hosts are visited.
8. Calculate and display the total cost of the broadcast tree.
9. Stop the program.

PROGRAM:

```
#include <stdio.h>

#define MAX 20

int main()
{
    int n, i, j, k;
    int cost[MAX][MAX];
    int visited[MAX] = {0};
    int min, u, v, total = 0;

    printf("Enter number of hosts: ");
    scanf("%d", &n);

    if (n < 1 || n > MAX)
    {
        printf("Invalid number of hosts.\n");
        return 0;
    }

    printf("Enter cost matrix (0 for no connection):\n");

    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &cost[i][j]);
        }
    }
}
```

```

visited[0] = 1;

printf("\nBroadcast Tree:\n");
printf("Root Host: Host 1\n");

for (i = 0; i < n - 1; i++)
{
    min = 999;
    u = -1;
    v = -1;

    for (j = 0; j < n; j++)
    {
        if (visited[j] == 1)
        {
            for (k = 0; k < n; k++)
            {
                if (visited[k] == 0 &&
                    cost[j][k] != 0 &&
                    cost[j][k] < min)
                {
                    min = cost[j][k];
                    u = j;
                    v = k;
                }
            }
        }
    }

    if (v == -1)
    {
        printf("All hosts are not connected.\n");
        return 0;
    }

    visited[v] = 1;
    total = total + min;

    printf("Host %d -> Host %d, Cost = %d\n",
           u + 1, v + 1, min);
}

printf("Total Cost of Broadcast Tree = %d\n", total);

return 0;
}

```

OUTPUT:

Enter number of hosts: 4

Enter cost matrix (0 for no connection):

0 2 4 0

2 0 1 5

4 1 0 3
0 5 3 0

Broadcast Tree:

Root Host: Host 1

Host 1 -> Host 2, Cost = 2

Host 2 -> Host 3, Cost = 1

Host 3 -> Host 4, Cost = 3

Total Cost of Broadcast Tree = 6